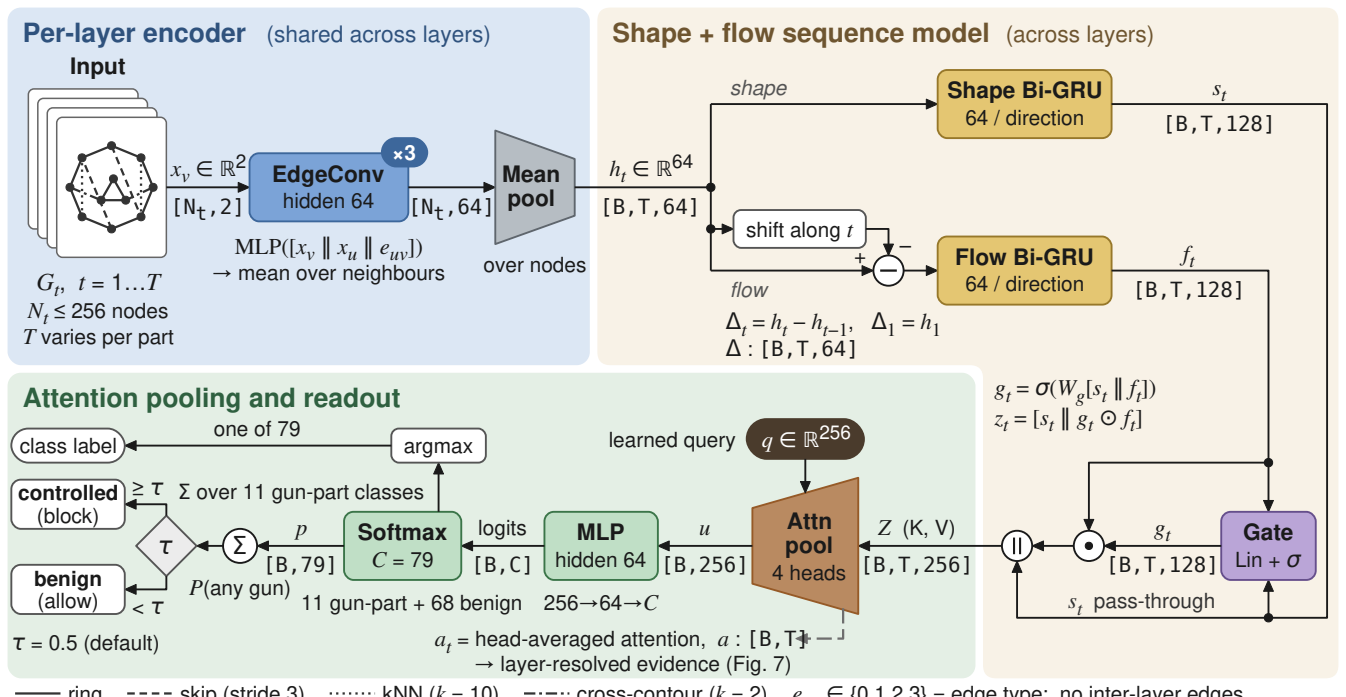

Per-layer encoder (shared across layers)
Input
G_t, $t = 1 \ldots T$
$N_t \le 256$ nodes
T varies per part
$x_v \in \mathbb{R}^2$
[N_t,2]
EdgeConv
hidden 64
×3
[N_t,64]
MLP([$x_v \parallel x_u \parallel e_{uv}$])
→ mean over neighbours
Mean pool
over nodes
$h_t \in \mathbb{R}^{64}$
[B,T,64]
Shape + flow sequence model (across layers)
shape
Shape Bi-GRU
64 / direction
s_t
[B,T,128]
shift along t
flow
Flow Bi-GRU
64 / direction
f_t
[B,T,128]
$\Delta_t = h_t - h_{t-1}$, $\Delta_1 = h_1$
Δ : [B,T,64]
$g_t = \sigma(W_g[s_t \parallel f_t])$
$z_t = [s_t \parallel g_t \odot f_t]$
Attention pooling and readout
learned query
$q \in \mathbb{R}^{256}$
Attn pool
4 heads
Z (K, V)
[B,T,256]
u
[B,256]
MLP
hidden 64
logits
[B,C]
Softmax
$C = 79$
p
[B,79]
Σ over 11 gun-part classes
argmax
one of 79
class label
controlled
(block)
benign
(allow)
≥ τ
< τ
τ
P(any gun)
11 gun-part + 68 benign
256→64→C
τ = 0.5 (default)
a_t = head-averaged attention, a : [B,T]
→ layer-resolved evidence (Fig. 7)
g_t
[B,T,128]
Gate
Lin + σ
s_t pass-through
ring
skip (stride 3)
kNN (k = 10)
cross-contour (k = 2)
$e_{uv} \in \{0,1,2,3\}$ = edge type; no inter-layer edges
subtract
elementwise product
concatenate
sum
side output (a_t)
*BatchNorm + ReLU after each EdgeConv, LayerNorm on h_t, dropout, and padding masks omitted.